Digital Electronics Question and Answers

Memory Devices-3

Topic 12

1.) Random Access Memory

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Digital Electronics Question & Answers

1.) Random Access Memory

1. What is the access time?
a) The time taken to move a stored word from one bit to other bits after applying the address bits
b) The time taken to write a word after applying the address bits
c) The time taken to read a stored word after applying the address bits
d) The time taken to erase a stored word after applying the address bits
Answer: c
Explanation: The access time is the time taken to read a stored word after applying the address bits in a MOS EPROM. It is the time required to fetch data from the memory.

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2. What are the typical values of tOE?
a) 10 to 20 ns for bipolar
b) 25 to 100 ns for NMOS
c) 12 to 50 ns for CMOS
d) All of the Mentioned

Answer: d
Explanation: The access time is the time taken to read a stored word after applying the address bits in a MOS EPROM. It is the time required to fetch data from the memory. The typical values of tOE (i.e. access time) are 10 to 20 ns for bipolar, 25 to 100 ns for NMOS and 12 to 50 ns for CMOS.

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3. Which of the following is not a type of memory?
a) RAM
b) FPROM
c) EEPROM
d) ROM
Answer: c
Explanation: EEPROM (Electrical Erasable Programmable ROM) is not a type of memory because it is used for erasing purpose only. Through EEPROM, data can be erased electrically, thereby consuming less time.

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4. The chip by which both the operation of read and write is performed ___________
a) RAM
b) ROM
c) PROM
d) EPROM

Answer: a

Explanation: A Random Access Memory (RAM) is a volatile chip memory in which both the read and write operations can be performed. Since it is volatile, therefore it stores data as long as power is on.

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5. RAM is also known as ___________

a) RWM

b) MBR

c) MAR

d) ROM

Answer: a

Explanation: A Random Access Memory (RAM) is a volatile chip memory in which both the read and write operations can be performed. Since it is volatile, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory).

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6. If a RAM chip has n address input lines then it can access memory locations upto ___________

a) $2^{(n-1)}$

b) $2^{(n+1)}$

c) 2^n

d) 2^{2n}

Answer: c

Explanation: RAM is a volatile memory, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory). If a RAM chip has n address input lines then it can access memory locations upto 2^n.

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7. The n-bit address is placed in the ___________

a) MBR

b) MAR

c) RAM

d) ROM

Answer: b

Explanation: The n-bit address is placed in the Memory Address Register (MAR) to select one of 2^n memory locations. It stores the address of the instruction which is to be executed next.

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8. Which of the following control signals are selected for read and write operations in a RAM?

a) Data buffer

b) Chip select

c) Read and write

d) Memory

Answer: c

Explanation: Read and write are control signals that are used to enable memory for read and write operations respectively.

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9. Computers invariably use RAM for ___________

a) High complexity

b) High resolution

c) High speed main memory

d) High flexibility

Answer: c

Explanation: RAM is a volatile memory, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory). Computers invariably use RAM for their high high-speed main memory and then use backup or slower-speed memories to hold auxiliary data.

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10. How many types of RAMs are?

a) 2

b) 3

c) 4

d) 5

Answer: a

Explanation: There are two types of RAM and these are static and dynamic. Static RAM(SRAM) is faster than dynamic RAM(DRAM) as the access time for DRAM is more compared to that of SRAM.

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11. Static RAM employs ___________

a) BJT or MOSFET

b) FET or JFET

c) Capacitor or BJT

d) BJT or MOS

Answer: d

Explanation: Static RAM employs bipolar or MOS flip-flops because both the semiconductor has storing capacity. Thus, it's access time is less and it is faster in operation.

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12. Dynamic RAM employs ___________

a) Capacitor or MOSFET

b) FET or JFET

c) Capacitor or BJT

d) BJT or MOS

Answer: a

Explanation: Dynamic RAM employs a capacitor or MOSFET. Thus, it's access time is more and it is slower in operation.

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13. Which one of the following is volatile in nature?

a) ROM

b) EROM

c) PROM

d) RAM

Answer: d

Explanation: RAM is a volatile memory, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory). RAMs are volatile because the stored data will be lost once the d.c. power applied to the flip-flops is removed.

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14. The magnetic core memories have been replaced by semiconductor RAMs, why?

a) Semiconductor RAMs are highly flexible

b) Semiconductor RAMs have highest storing capacity

c) Semiconductor RAMs are smaller in size

d) All of the Mentioned

Answer: d

Explanation: RAM is a volatile memory, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory). The magnetic core memories have been replaced by semiconductor RAMs because of smaller in size, high storing capacity as well as flexibility.

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15. The data written in flip-flop remains stored as long as ___________

a) D.C. power is supplied

b) D.C. power is removed

c) A.C. power is supplied

d) A.C. power is removed

Answer: a

Explanation: Since flip-flops are made up of semiconductor materials. So, it can't accept A.C. source and the data written in flip-flop remains stored as long as the dc power is maintained.

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16. The memory capacity of a static RAM varies from _____________

a) 32 bit to 64 bit

b) 64 bit to 1024 bit

c) 64 bit to 1 Mega bit
d) 512 bit to 1 Mega bit
Answer: c
Explanation: Static RAM(SRAM) is faster than dynamic RAM(DRAM) as the access time for DRAM is more compared to that of SRAM. The memory capacity of a static RAM varies from 64 bits to 1 Mega bit.

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17. The input data bit is written into the cell by setting ____________
a) The flip-flop for 1
b) Resetting the flip-flop
c) The flip-flop for HIGH
d) Both the flip-flop for 1 and resetting the flip-flop
Answer: d
Explanation: The input data bit (1 or 0) is written into the cell by setting the flip-flop for 1 and the resetting the flip-flop for a 0 when the R/W' line is low, R/W is active-low pin.

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18. When the READ/(WRITE)' line is HIGH then the flip-flop is ____________
a) Activated
b) Deactivated
c) Unaffected
d) Both activated and deactivated

Answer: c
Explanation: When the READ/(WRITE)' line is HIGH then the flip-flop is unaffected since it's an active low pin. It means that the stored bit (data) is gated to the data out line.

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19. The flip-flop in static memory cell can be constructed using ____________
a) Capacitor or MOSFET
b) FET or JFET
c) Capacitor or BJT
d) BJT or MOSFET
Answer: d
Explanation: The flip-flop in the static memory cell can be constructed using Bipolar Junction Transistor (BJT) and MOSFETs because of it's storing capability. Also, it's access time is less and it is faster in operation.

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20. Which of the following IC is TTL based static RAM?
a) IC 7488
b) IC 7489
c) IC 7487
d) IC 2114

View Answer

Answer: b
Explanation: The flip-flop in the static memory cell can be constructed using Bipolar Junction Transistor (BJT) and MOSFETs because of it's storing capability. Also, it's access time is less and it is faster in operation. In IC 7489, TTL is used which is static RAM.

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21. IC 7489 is of _____________
a) 32 bit
b) 64 bit
c) 512 bit
d) 1024 bit
Answer: b
Explanation: The arrangement of IC 7489 is 16 * 4 = 64 bits. Thus, it has 16 columns and 4 rows.

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22. Data is written in IC 7489 through _____________
a) Chip select
b) Enable
c) Data input
d) Memory enable

Answer: c
Explanation: Data can be written into the memory via the data inputs by supplying an address to the SELECT inputs and holding both the memory enable and write enable LOW.

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23. The first practical form of Random Access Memory (RAM) was the _____________
a) Cathode tube
b) Data tube
c) Memory tube
d) Select tube
Answer: c
Explanation: Data can be written into the memory via the data inputs by supplying an address to the SELECT inputs and holding both the memory enable and write enable LOW since it's an active-low pin.

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24. Magnetic-core memory was invented in _____________
a) 1946
b) 1948
c) 1947
d) 1945

Answer: c

Explanation: Magnetic-core memory was invented in 1947 and developed up until the mid-1970s. It became a widespread form of random-access memory, relying on an array of magnetized rings. RAM is a volatile memory, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory).

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25. Which of the following RAM is volatile in nature?
a) SRAM
b) DRAM
c) EEPROM
d) Both SRAM and DRAM

Answer: d

Explanation: RAM is a volatile memory, therefore it stores data as long as power is on. RAM is also known as RWM (i.e. Read Write Memory). Both static and dynamic RAM are considered volatile, as their state is lost or reset when power is removed from the system.

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26. Which one of the following IC is of 4KB?
a) IC 7488
b) IC 7489
c) IC 7487
d) IC 2114

Answer: d

Explanation: IC 2114 is of 4KB and it is a static RAM. Both static and dynamic RAM are considered volatile, as their state is lost or reset when power is removed from the system.

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27. Which of the following IC implements the static MOS RAM?
a) TMS 4015
b) TMS 4014
c) TMS 4016
d) TMS 2114

Answer: c

Explanation: The IC TMS 4016 is a 2KB static MOS RAM. There are eleven address inputs available to select the 2014 locations.

---

28. What types of arrangements a TMS 4016 has?
a) 1024 * 4
b) 1024 * 8
c) 2048 * 4
d) 2048 * 8

Answer: d

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Explanation: The IC TMS 4016 is a 2KB static MOS RAM. There are eleven address inputs available to select the 2014 locations.
Therefore, arrangements in TMS 4016 = 2048 * 8 = 2KB.

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29. How many address inputs are available in TMS 4016?
a) 10
b) 9
c) 12
d) 11
Answer: d
Explanation: The IC TMS 4016 is a 2KB static MOS RAM. There are eleven address inputs available to select the 2014 locations.

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30. In TMS 4016, the data in/data out are _______________
a) Unidirectional
b) Parallel
c) Serial
d) Bidirectional

Answer: d
Explanation: In TMS 4016, the data in/data out are bidirectional terminal. It means that the input/output can be transmitted or received through the same terminal.

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31. Dynamic RAM is more preferable than static RAM, why?
a) DRAM is of the lowest cost, lowest density
b) DRAM is of the highest cost, reduced size
c) DRAM is of the lowest cost, highest density
d) DRAM is more flexible and lowest storage capacity
Answer: c
Explanation: The Dynamic Random Access Memory is the lowest cost, highest density random access memory available. Nowadays, computers use DRAM for main memory. However, it's access time is more compared to SRAM.

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32. The memory size of DRAM is _____________
a) 1 to 100 MB
b) 512 to 1024 MB
c) 64 to 512 MB
d) 16 to 256 MB
Answer: d
Explanation: The Dynamic Random Access Memory is the lowest cost, highest density random access memory available. Nowadays, computers use DRAM for main memory.

However, it's access time is more compared to SRAM. The memory size of DRAM lies between 16 to 256 MB.

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33. The DRAM stores its binary information on ___________
a) MOSFET
b) Transistor
c) Capacitor
d) BJT

Answer: c
Explanation: Capacitor has high storing capability only, so DRAM stores its binary information in the form of electric charges on capacitors. However, DRAM takes more time time to access data.

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34. Most modern operating systems employ a method of extending RAM capacity, known as ___________
a) Magnetic memory
b) Virtual memory
c) Storage memory
d) Static memory

Answer: b
Explanation: Most modern operating systems employ a method of extending RAM capacity, known as virtual memory. Virtual memory is seen as part of main memory but is actually a secondary memory.

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35. DRAM uses of integrated MOS capacitors as ________ instead of a flip-flop.
a) Storage cell
b) Memory cell
c) Dynamic cell
d) Static cell
Answer: b
Explanation: DRAM uses of integrated MOS capacitors as memory cell instead of a flip-flop. The advantage of this cell is that it allows very large memory arrays to be constructed on a chip at a lower cost per bit than in static memories.

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36. What is the disadvantage of MOS capacitor in DRAM?
a) It can't hold the data till a long period
b) It doesn't holds the charge till a long period
c) It is highly densed
d) It is not flexible

Answer: b

Explanation: The disadvantage of MOS capacitor in DRAM is that it can't hold the stored charge over a long period of time and it has to be refreshed every few millisecond. Thus, DRAM is slow in operation.

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37. The dynamic RAM offers __________

a) High power consumption, large storage capacity

b) Reduced power consumption, large storage capacity

c) Reduced power consumption, short storage capacity

d) High power consumption, short storage capacity

Answer: b

Explanation: The dynamic RAM offers reduced power consumption and large storage capacity in a single memory chip. With the availability of such high packing density memory ICs, the capacity of memory will continue to grow. However, it's access time is more and thus operation is slow.

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38. The main memory of a PC is made of __________

a) Cache

b) Dynamic RAM

c) Static RAM

d) Both cache and dynamic RAM

Answer: d

Explanation: The main memory of a PC is made of cache and DRAM. DRAM offers reduced power consumption and large storage capacity in a single memory chip.

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39. Virtual memory consists of __________

a) SRAM

b) DRAM

c) Magnetic memory

d) Main Memory

Answer: a

Explanation: Most modern operating systems employ a method of extending RAM capacity, known as virtual memory which consists of SRAM.

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40. Dynamic RAM is used as main memory in a computer system as __________

a) It has a lower cell density

b) It needs refreshing circuitry

c) Consumes less power

d) Has higher speed

Answer: d

Explanation: Dynamic RAM is used as main memory in a computer system as it has a higher speed due to the presence of MOSFET technology. However, it's access time is more compared to SRAM and operation is slow.

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41. Cache memory acts between ___________
a) RAM and ROM
b) CPU and RAM
c) CPU and Hard Disk
d) CPU and ROM

Answer: b

Explanation: In a computer, cache memory acts between CPU and RAM.

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42. Which characteristic of RAM memory makes it not suitable for permanent storage?
a) Unreliable
b) Too slow
c) Too bulky
d) It is volatile

Answer: d

Explanation: RAM is volatile. Therefore, it stores data only as long as the d.c power is on.

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43. Why do a DRAM employ the address multiplexing technique?
a) To reduce the number of memory locations
b) To increase the number of memory locations
c) To reduce the number of address lines
d) To increase the number of address lines

Answer: c

Explanation: A Dynamic RAM usually employs a technique called address multiplexing to reduce the number of address lines and thus the number of input/output pins on the IC package.

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44. An address multiplexing in DRAM is of ______ bits.
a) 10240
b) 15289
c) 16384
d) 17654

Answer: c

Explanation: A Dynamic RAM usually employs a technique called address multiplexing to reduce the number of address lines and thus the number of input/output pins on the IC package. Address multiplexing has 2^{14} = 16384 bits.

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45. What is a sense amplifier?
a) It is an amplifier which converts ac current into dc current
b) It is an amplifier which lowers the input voltage
c) It is an amplifier which increases the input voltage
d) It is an amplifier which converts the low voltage to a sufficient voltage
Answer: d
Explanation: A sense amplifier for each column is necessary to convert from the low voltage and low energy to a sufficient level on the I/O data line.

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46. DRAM is fabricated by using IC __________
a) 2114
b) 7489
c) 4116
d) 2776
Answer: c
Explanation: DRAM is Dynamic RAM which takes more access time compared to SRAM and is thus, slower in operation comparatively. Although, in general it offers high speed and is used in most computers nowadays. DRAM is fabricated by using IC 4116.

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47. IC 4116 is of ______ storage memory.
a) 16 KB
b) 32 KB
c) 64 MB
d) 2 KB
Answer: a
Explanation: IC 4116 is a DRAM of 16 KB storage memory. It requires three supply voltages (+5V, -5V, and +12V) to operate the IC unit.

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48. How many supply voltage IC 4116 requires to operate the IC unit?
a) 3
b) 2
c) 1
d) 4

Answer: a
Explanation: IC 4116 is a DRAM of 16 KB storage memory. It requires three supply voltages (+5V, -5V, and +12V) to operate the IC unit.

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49. The full form of PSRAM is __________
a) Plugged Static RAM
b) Plugged Stored RAM

c) Pseudo Stored RAM
d) Pseudo Static RAM

Answer: d
Explanation: The full form of PSRAM is Pseudo Static RAM. It is a dynamic RAM which is implemented as a SRAM.

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50. Pseudo static RAM is a ___________
a) Static RAM
b) Dynamic RAM
c) Cache
d) ROM
Answer: b
Explanation: The full form of PSRAM is Pseudo Static RAM. It is a dynamic RAM having built-in fresh logic, which is implemented as an SRAM.

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51. When PSRAM is performing internal refresh ___________
a) The read operation is performed
b) The write operation is performed
c) It can not be accessed for read or write
d) The voltage goes HIGH
Answer: c
Explanation: The full form of PSRAM is Pseudo Static RAM. It is a dynamic RAM having built-in fresh logic, which is implemented as a SRAM. So, it can not be accessed for read or write during the refresh operation.

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52. RAMs are utilized in the computer as ___________
a) Scratch-pad
b) Buffer
c) Main memory
d) All of the Mentioned

Answer: d
Explanation: RAMs are utilized in the computer as a scratch-pad, buffer and main memories. These are the applications of RAMs. Mostly, these RAMs are DRAMs as they provide high speed.

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53. The advantages of RAMs are ___________
a) Non destructive read out
b) Fast operating speed
c) Low power dissipation

d) All of the Mentioned

Answer: d

Explanation: The advantages of RAM are Non-destructive read out, Fast operating speed and Low power dissipation.

---

54. Which one is more economical?

a) ROM

b) RAM

c) EROM

d) PROM

Answer: b

Explanation: RAM is more economical than ROM because MOS memories are more economical than the magnetic core for small and medium sized systems.

---

55. Which one is self-compatible?

a) ROM

b) RAM

c) EROM

d) PROM

Answer: b

Explanation: As semiconductor memories enjoy common interface and technology between sensing and decoding circuitry and the storage element itself, so RAMs are self-compatible. Also, they provide high speed and fast operation.

---

56. The memory which is used for storing programs and data currently being processed by the CPU is called ___________

a) PROM

b) Main Memory

c) Non-volatile memory

d) Mass memory

Answer: a

Explanation: PROM has the capability to store the data due to the presence of MOSFET which is processed by the CPU. It is one-time programmable by the user.

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57. CD-ROM is a ___________

a) Memory register

b) Magnetic memory

c) Semiconductor memory

d) Non-volatile memory

Answer: d

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Explanation: CD-ROM is a non-volatile memory. Once a program is uploaded in it then it can't be erasable. Thus, it stores the data permanently.

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58. A place which is used as storage location in a computer __________
a) A bit
b) A record
c) An address
d) A byte

Answer: c
Explanation: A storage location of a computer is an address/memory location, used to store instructions and data.

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59. Which of the following is not a primary storage device?
a) Optical disk
b) Magnetic tape
c) Magnetic disk
d) RAM
Answer: d
Explanation: RAM (i.e. Random Access Memory) is not a primary storage device.

2.) Programmable Logic Array

1. What is memory decoding?
a) The process of Memory IC used in a digital system is overloaded with data
b) The process of Memory IC used in a digital system is selected for the range of address assigned
c) The process of Memory IC used in a digital system is selected for the range of data assigned
d) The process of Memory IC used in a digital system is overloaded with data allocated in memory cell
Answer: b
Explanation: The Memory IC used in a digital system is selected or enabled only for the range of addresses assigned to it and this process is called memory decoding. It decodes the memory to be selected for a specific address.

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2. The first step in the design of memory decoder is __________
a) Selection of a EPROM
b) Selection of a RAM
c) Address assignment
d) Data insertion

Answer: c

Explanation: Memory decoder decodes the memory to be selected for a specific address. The first step in the design of memory decoder is address assignment in non-overlapped manner.

3. How many address bits are required to select memory location in the Memory decoder?
a) 4 KB
b) 8 KB
c) 12 KB
d) 16 KB

Answer: c

Explanation: Memory decoder decodes the memory to be selected for a specific address. Since the given EPROM and RAM are of 4 KB (4 * 1024 = 4096) capacity, it requires 12 address bit to select one of the 4096 memory locations.

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4. How memory expansion is done?
a) By increasing the supply voltage of the Memory ICs
b) By decreasing the supply voltage of the Memory ICs
c) By connecting Memory ICs together
d) By separating Memory ICs

Answer: c

Explanation: Memory ICs can be connected together to expand the number of memory words or the number of bits per word.

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5. IC 4116 is organised as __________
a) 512 * 4
b) 16 * 1
c) 32 * 4
d) 64 * 2

Answer: b

Explanation: IC 4116 is organised as 16 * 1 K which has capability to store 16 KB.

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6. To construct 16K * 4-bit memory, how many 4116 ICs are required?
a) 1
b) 2
c) 3
d) 4

Answer: d

Explanation: Since, IC 4116 is organised as 16K * 1, which can store about 16KB data. So, four ICs are required for 16K * 4 memory implementation.

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7. How many 1024 * 1 RAM chips are required to construct a 1024 * 8 memory system?
a) 4
b) 6
c) 8
d) 12
Answer: c
Explanation: One 1024 * 1 RAM chips is of 1-bit. SO, for construction of 1024 * 8 RAM chip of 8-bits, it will require 8 chips.

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8. How many 16K * 4 RAMs are required to achieve a memory with a capacity of 64K and a word length of 8 bits?
a) 2
b) 4
c) 6
d) 8
Answer: d
Explanation: 16K * 4 = 64K RAM is of 64K. Therefore, for a word of length 8-bits, 64 * 8 = 512K RAM required. Thus, number of 16K * 4 RAMs = 512/64 = 8.

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9. The full form of PLD is __________
a) Programmable Load Devices
b) Programmable Logic Data
c) Programmable Logic Devices
d) Programmable Loaded Devices

Answer: c
Explanation: The full form of PLD is Programmable Logic Devices. It is a collection of gates, flip-flops and registers on a single chip.

---

10. PLD contains a large number of __________
a) Flip-flops
b) Gates
c) Registers
d) All of the Mentioned
Answer: d
Explanation: Programmable Logic Devices is a collection of a large number of gates, flip-flops, registers that are interconnected on the chip. Thus, it is used for designing logic circuits.

---

11. Logic circuits can also be designed using __________
a) RAM

b) ROM
c) PLD
d) PLA
Answer: c
Explanation: Programmable Logic Devices is a collection of large number of gates, flip-flops, registers that are interconnected on the chip. Thus, it is used for designing logic circuits.

---

12. In PLD, there are provisions to perform interconnections of the gates internally, because of __________
a) High reliability
b) High conductivity
c) The desired logic implementation
d) The desired output
Answer: c
Explanation: Programmable Logic Devices is a collection of a large number of gates, flip-flops, registers that are interconnected on the chip. In PLD, there are provisions to perform interconnections of the gates internally so that the desired logic can be implemented.

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13. Why antifuses are implemented in a PLD?
a) To protect from high voltage
b) To increase the memory
c) To implement the programmes
d) As a switching devices

Answer: c
Explanation: Programmable Logic Devices is a collection of a large number of gates, flip-flops, registers that are interconnected on the chip. Programming is accomplished by using antifuses in a PLD and it is fabricated at the cross points of the gates.

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14. How many types of PLD is?
a) 2
b) 3
c) 4
d) 5

Answer: a
Explanation: There are two types of PLD, viz., devices with fixed architecture and devices with a flexible architecture. The main categories of PLDs are PROM, PAL and PLA.

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15. PLA refers to __________
a) Programmable Loaded Array
b) Programmable Array Logic
c) Programmable Logic Array
d) Programmed Array Logic

Answer: c
Explanation: PLA refers to Programmable Logic Array. It is a type of PLD having programmable AND and OR gates.

3.) Programmable Array Logic

1. The inputs in the PLD is given through _____________
a) NAND gates
b) OR gates
c) NOR gates
d) AND gates

Answer: d
Explanation: The inputs in the PLD is given through AND gate followed by inverting & non-inverting buffer. PLDs are Programmable Logic Devices consisting of logic gates, flip-flops and registers connected together on a single chip. Thus, it can be categorised into PROM, PAL and PLA.

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2. PAL refers to _____________
a) Programmable Array Loaded
b) Programmable Logic Array
c) Programmable Array Logic
d) Programmable AND Logic

Answer: c
Explanation: PAL refers to Programmable Array Logic consisting of programmable AND gates and fixed OR gates.

---

3. Outputs of the AND gate in PLD is known as _____________
a) Input lines
b) Output lines
c) Strobe lines
d) Control lines
Answer: b
Explanation: Outputs of the AND gate in PLD is known as output lines.

---

4. PLA contains _____________
a) AND and OR arrays
b) NAND and OR arrays
c) NOT and AND arrays
d) NOR and OR arrays
Answer: a
Explanation: Programmable Logic Array is a type of fixed architecture logic devices with programmable AND gates followed by programmable OR gates. It is a kind of PLD.

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5. PLA is used to implement _____________
a) A complex sequential circuit
b) A simple sequential circuit
c) A complex combinational circuit
d) A simple combinational circuit
Answer: c
Explanation: Since, PLA is a combination of programmable AND and OR gates. So, it is used to implement complex combinational circuit.

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6. A PLA is similar to a ROM in concept except that _____________
a) It hasn't capability to read only
b) It hasn't capability to read or write operation
c) It doesn't provide full decoding to the variables
d) It hasn't capability to write only

Answer: c
Explanation: A PLA is similar to a ROM in concept except that it doesn't provide full decoding to the variables and doesn't generate all the minterms as in the ROM. Programmable Logic Array is a type of fixed architecture logic devices with programmable AND gates followed by programmable OR gates. It is a kind of PLD.

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7. For programmable logic functions, which type of PLD should be used?
a) PLA
b) PAL
c) CPLD
d) SLD
Answer: b
Explanation: Since PAL consists of programmable AND gates and fixed OR gates and also circuitry working is less.

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8. The complex programmable logic device contains several PLD blocks and ___________
a) A language compiler

b) AND/OR arrays
c) Global interconnection matrix
d) Field-programmable switches
Answer: c
Explanation: The complex programmable logic device contains several PLD blocks and a global interconnection matrix by which it communicates through several devices. It is also known as Field-Programmable Gate Arrays (FPGAs).

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9. Which type of device FPGA are?
a) SLD
b) SROM
c) EPROM
d) PLD
Answer: d
Explanation: Field-Programmable Gate Arrays (FPGAs) are reprogrammable silicon chips. In contrast to processors that you find in your PC, programming an FPGA rewires the chip itself to implement your functionality rather than run a software application. Thus, FPGAs are PLD devices.

---

10. The difference between a PAL & a PLA is ____________
a) PALs and PLAs are the same thing
b) The PLA has a programmable OR plane and a programmable AND plane, while the PAL only has a programmable AND plane
c) The PAL has a programmable OR plane and a programmable AND plane, while the PLA only has a programmable AND plane
d) The PAL has more possible product terms than the PLA
Answer: b
Explanation: The main difference between a PAL & PLA is that PLA has a programmable OR plane and a programmable AND plane, while the PAL only has a programmable AND plane and a fixed OR plane.

---

11. If a PAL has been programmed once ____________
a) Its logic capacity is lost
b) Its outputs are only active HIGH
c) Its outputs are only active LOW
d) It cannot be reprogrammed
Answer: d
Explanation: PAL only has a programmable AND plane and a fixed OR plane. Since, PAL is dynamic in nature. So, it can’t be reprogrammed.

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12. The FPGA refers to _____________
a) First programmable Gate Array
b) Field Programmable Gate Array
c) First Program Gate Array
d) Field Program Gate Array

Answer: b

Explanation: The FPGA refers to Field Programmable Gate Array. Field-Programmable Gate Arrays (FPGAs) are reprogrammable silicon chips. In contrast to processors that you find in your PC, programming an FPGA rewires the chip itself to implement your functionality rather than run a software application. Thus, FPGAs are PLD devices.

---

13. The full form of VLSI is _____________
a) Very Long Single Integration
b) Very Least Scale Integration
c) Very Large Scale Integration
d) Very Long Scale Integration

Answer: c

Explanation: The full form of VLSI is Very Large Scale Integration in which FPGA is implemented.

---

14. In FPGA, vertical and horizontal directions are separated by _____________
a) A line
b) A channel
c) A strobe
d) A flip-flop

Answer: b

Explanation: The FPGA refers to Field Programmable Gate Array. Field-Programmable Gate Arrays (FPGAs) are reprogrammable silicon chips. Vertical and horizontal directions is separated by a channel in an FPGA which determines the location of the output.

---

15. Applications of PLAs are ______________
a) Registered PALs
b) Configurable PALs
c) PAL programming
d) All of the Mentioned

Answer: d

Explanation: Applications of PLAs are Registered PALs, Configurable PALs, and PAL programming and these are performed by using an extra flip-flop with PAL.

4.) Introduction to Hardware Description Language

1. The full form of HDL is ________________

a) Higher Descriptive Language

b) Higher Definition Language

c) Hardware Description Language

d) High Descriptive Language

Answer: c

Explanation: The full form of HDL is Hardware Description Language.

---

2. The full form of VHDL is ____________

a) Very High Descriptive Language

b) Verilog Hardware Description Language

c) Variable Definition Language

d) None of the Mentioned

Answer: b

Explanation: The full form of VHDL is Verilog Hardware Description Language.

---

3. VHSIC stands for ____________

a) Very High Speed Integrated Circuits

b) Very Higher Speed Integration Circuits

c) Variable High Speed Integrated Circuits

d) Variable Higher Speed Integration Circuits

Answer: a

Explanation: VHSIC stands for Very High Speed Integrated Circuits.

---

4. VHDL is being used for ____________

a) Documentation

b) Verification

c) Synthesis of large digital design

d) All of the Mentioned

Answer: d

Explanation: The full form of VHDL is Verilog Hardware Description Language. The acronym of VHDL itself captures the entire theme of the language and it describes the hardware in the same manner as does the schematic. So, it is used as documentation, verification and synthesis of large digital designs.

---

5. The use of VHDL can be done in _____ ways.

a) 2

b) 3

c) 4
d) 5
Answer: b
Explanation: The VHDL has three coding styles are: (i) data flow, (ii) structural, (iii) behavioural.

---

6. At high frequencies when the sampling interval is too long in a frequency counter ______________

a) The counter works fine
b) The counter undercounts the frequency
c) The measurement is less precise
d) The counter overflows
Answer: d
Explanation: Let the sampling time be 1 sec. This means the counter will count the number of pulses from the unknown signal for 1sec duration and would display it after 1 sec. thus if the signal is of 800 Hz, at the end of 1 sec, counter would have counted up to 800. Thus, in case of high frequencies and high sampling time, counter might count beyond its limit and overflows.

---

7. The output frequency related to the sampling interval of a frequency counter as ______________

a) Directly with the sampling interval
b) Inversely with the sampling interval
c) More precision with longer sampling interval
d) Less precision with longer sampling interval
Answer: c
Explanation: Sampling interval means a particular frequency range in which the device operates correctly. Thus, more precision is produced with longer sampling interval.

---

8. In an HDL application of a stepper motor, what is done next after an up/down counter is built?
a) Build the sequencer
b) Test it on a simulator
c) Test the decoder
d) Design an intermediate integer variable

Answer: b
Explanation: Simulator is a software which is used in the testing of the stepper motor using up/down counter.

---

9. In a digital clock application, the basic frequency must be divided down as ______________

a) 1 Hz
b) 60 Hz
c) 100 Hz
d) 1000 Hz

Answer: a

Explanation: Minimum count is 1 sec and time = 1/freq. So, t = 1/1 = 1Hz.

---

10. What does the data signal do in the keypad application?

a) The row and column encoded data
b) The ring encoded data
c) The freeze locator data
d) The ring counter data

Answer: a

Explanation: The data signal arranges the information with the help of data flow in row and column manner. It encodes the data to be sent.

---

11. When a key is pressed, what does the ring counter in the HDL keypad application do?

a) Count to find the row
b) Freeze
c) Count to find the column
d) Start the D flip-flop

Answer: a

Explanation: The data signal arranges the information with the help of data flow in row and column manner. It encodes the data to be sent. When a key is pressed the ring counter in the HDL scans the information provided by the user and counts to find the row.

---

12. A step which should be followed in project management is known as ______________

a) Overall definition
b) System documentation
c) Synthesis and testing
d) System integration

Answer: b

Explanation: System documentation is the second step of project management in which a result of the system is noted simultaneously.

13. In the keypad application, the preset state of the ring counter define ______________

a) The NANDing of the columns
b) The NANDing of the rows

c) The proper output of the column encoder
d) The proper output of the row encoder
Answer: d
Explanation: When a key is pressed the ring counter in the HDL scans the information provided by the user and counts to find the row. The preset state of the ring counter define the proper output of the row encoder.

---

14. A major block which is not a part of an HDL frequency counter _______________
a) Timing and control unit
b) Decoder/display
c) Display register
d) Bit shifter
Answer: d
Explanation: Bit shifter is part of a register in which bit shifting takes place bit-by-bit either left or right.

---

15. A stepper motor HDL application must include _______________
a) Sequencers and multiplexers
b) Types and bits
c) Counters and decoders
d) Variables and processes

Answer: c
Explanation: A stepper motor (also referred to as step or stepping motor) is an electromechanical device achieving mechanical movements through the conversion of electrical pulses. A stepper motor HDL application must include counters and decoders for position control. It is tested on the simulator.

5.) Multivibrators

1. A multivibrator is an electronic circuit used to implement ______________
a) Oscillator
b) Timer
c) Flip-flop
d) All of the Mentioned
Answer: d
Explanation: A multivibrator is an electronic circuit used to implement a variety of simple two-state systems and two state systems are an oscillator, timer, flip-flop. It produces an output when triggered.

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2. Multivibrators are characterized by ____________

a) Registers

b) Capacitors

c) Transistors

d) All of the Mentioned

Answer: d

Explanation: A multivibrator is an electronic circuit used to implement a variety of simple two-state systems and two state systems are an oscillator, timer, flip-flop. It produces an output when triggered. Multivibrators are characterized by amplifying devices (transistors) and cross coupled devices (registers, capacitors).

---

3. How many types of multivibrators are?

a) 2

b) 4

c) 5

d) 3

Answer: d

Explanation: There are three types of multivibrator circuits depending on the circuit operation: (i) Astable, (ii) Bistable, and (iii) Monostable. Astable multivibrator is internally triggered, whereas the other two types are externally triggered.

---

4. Astable multivibrator is ________ in any state.

a) Stable

b) Unstable

c) Saturated

d) Both Stable & Saturated

Answer: b

Explanation: Astable multivibrator, in which the circuit is not stable in either state i.e. it continually switches from one state to the other. It is internally triggered and does not have any stable state.

---

5. Monostable multivibrator is/has ________ state.

a) Stable

b) Unstable

c) One stable and another unstable

d) Independent

Answer: b

Explanation: Monostable multivibrator, in which one of the states is stable, but the other

state is unstable (transient). A trigger pulse causes the circuit to enter the unstable state. After entering the unstable state, the circuit will return to the stable state after a set time.

---

6. Bistable multivibrator is _________ in any state.
a) Stable
b) Unstable
c) Saturated
d) Independent

Answer: a
Explanation: Bistable multivibrator is a type of multivibrator having two stable states. The circuit is stable in either state. It can be flipped from one state to the other by an external trigger pulse.

---

7. Bistable circuit is also known as _____________
a) Latch
b) Gate
c) Flip-flop
d) Bidirectional circuit
Answer: c
Explanation: Bistable multivibrator circuit has the capability to store 1-bit of information. So, it is also known as flip-flop. Like Flip-flop, Bistable circuit also has 2 states.

---

8. Astable circuit acts as a/an _____________
a) Amplifier
b) Oscillator
c) Relaxation oscillator
d) Multiplexer

Answer: c
Explanation: Astable circuit continually switches from one state to the other. It keeps oscillating between unstable states. Hence, it functions as a relaxation oscillator.

---

9. In an astable multivibrator, the amplifying elements are _____________
a) FET
b) JFET
c) OP-AMP
d) All of the Mentioned
Answer: d
Explanation: Astable multivibrators are made with FET, JFET, OP-AMP or other types of

amplifier. Astable circuit continually switches from one state to the other. It keeps oscillating between unstable states.

---

10. Monostable multivibrator can also be termed as ____________

a) Full astable multivibrator
b) Half astable multivibrator
c) Half bistable multivibrator
d) Full bistable multivibrator

Answer: b

Explanation: Since, astable multivibrator is unstable and changes their states continually (It means that it can have both the states), whereas in monostable multivibrator, one of the states is stable, but the other state is unstable. SO Monostable Multivibrator is also known as Half-Astable Multivibrator.

---

11. The classic multivibrator circuit is known as ____________

a) Metal-coupled multivibrator
b) Plate-coupled multivibrator
c) Parallel-plate coupled multivibrator
d) Alternate-plate coupled multivibrator

Answer: b

Explanation: The classic multivibrator circuit (also called a plate-coupled multivibrator) is first described by Henri Abraham and Eugene Bloch.

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